MILITARY SPECIFICATION

HEATER, GUN (28 VOLTS DC), TYPE J-4A

This specification has been approved by the Departments of the Air Force and by the Bureau of Naval Weapons.

1. SCOPE

1.1 This specification covers one type of electrical gun heater for operation on 28 volts, direct current (dc), designated type J-4A.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-676	Boxes, Set-Up, Paperboard

Military

MIL-P-116	Preservation, Methods of
MIL-E-5557	Enamel; Heat-Resisting, Glyceryl-Phthalate, Black
MIL-C-5756	Cable or Wire, Power, Electric, Portable
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-P-16232	Phosphate Coatings, Heavy, Manganese or Zinc Base (for Ferrous Metals)
MIL-W-25038	Wire, Electrical, High Temperature and Fire Resistant, Aircraft
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of

FSC 1095

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3.6.1.3 Cord.- Each conductor shall consist of a 19-strand No. 18 AWG size wire covered with an insulation whose outside diameter (OD) is a maximum of 0.112 inch. The wire and insulation, except for the OD of the insulation, shall be in accordance with MIL-W-25038. Each finished conductor shall be made into a twisted two-conductor cable, the buildup and sheath complying with MIL-C-5756. It shall be fastened to the heater pad backing plate in a manner that will permit the cord to be brought out from the front or rear. The cord shall be securely fastened to the heater pad backing plate in a manner that will prevent breakage when subjected to vibrations encountered when mounted on a free-firing machine gun, and shall be terminated in a connector as specified on 53D4988.

3.7 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of MIL-D-70327.

3.8 Weight.- The overall weight shall not exceed 0.5 pound.

3.9 Finishes and protective coating.-

3.9.1 Corrosion.- The heater housing and all parts subject to corrosion shall be treated in accordance with MIL-P-16232, type M, class 1.

3.9.2 Exterior surface.- The heater shall be finished with one coat of instrument black enamel conforming to MIL-E-5557, type IV.

3.10 Performance.-

3.10.1 Temperature rise.- The heater shall be capable of producing a minimum temperature rise, as specified in table I, when mounted on a test fixture as shown on 44B25260, and when operated in still air on the ground with an ambient temperature of 70° ±5° F.

TABLE I. Temperature

Time (minutes)	Temperature rise (°F)
30	70
60	130
90	165
120	180

3.10.2 Electrical requirements.- The electrical energy required for operation consistent with the requirements of the temperature rise shall be 100 watts ±5 watts at 28 volts dc.

3.11 Identification of product.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130, and shall be applied by means of a steel stamp.

3.12 Installation instructions.- The contractor shall furnish with each heater, one printed copy of installation instructions. Prior to printing, two copies shall be furnished to the procuring activity for approval. The instructions shall be arranged to require only one sheet of paper, no larger than 8-1/2 by 11 inches. Each set of instructions shall contain the following data and a copy of figure 1:

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(a) Prior to installing the heater on the gun, the cable shall be placed in the desired position in the housing (running fore or aft) and safety-wired in position. Because of hardening of the rubber insulation that occurs after continued use, where the cable enters the heating element assembly, the position of the cable in the housing should not be changed after the heater has been in service.

(b) To install gun heater:

(1) The lower edge of the side of the heater having three holes shall be hooked over the lower edge of the gun cover at the positions dictated by the particular aircraft installation.

(2) The heater shall be flexed to clear opposite edge of cover and pressed down to snap into place.

(c) After installation, the heater should be examined to ensure that the heating pad is in good contact with the side of the receiver. The receiver cover should then be opened and closed to ensure that the heater is securely installed on the cover.

(d) To remove gun heater, steps (b) (1) and (b) (2) above shall be followed in reverse.

(e) This heater can be installed in the positions as shown below with the electrical cable being brough out either side of the heater.

IMPORTANT

To ensure longer life of the heater and prevent deterioration of the copper heating surface, the heater should not be operated when off the gun or when gun cover is raised.

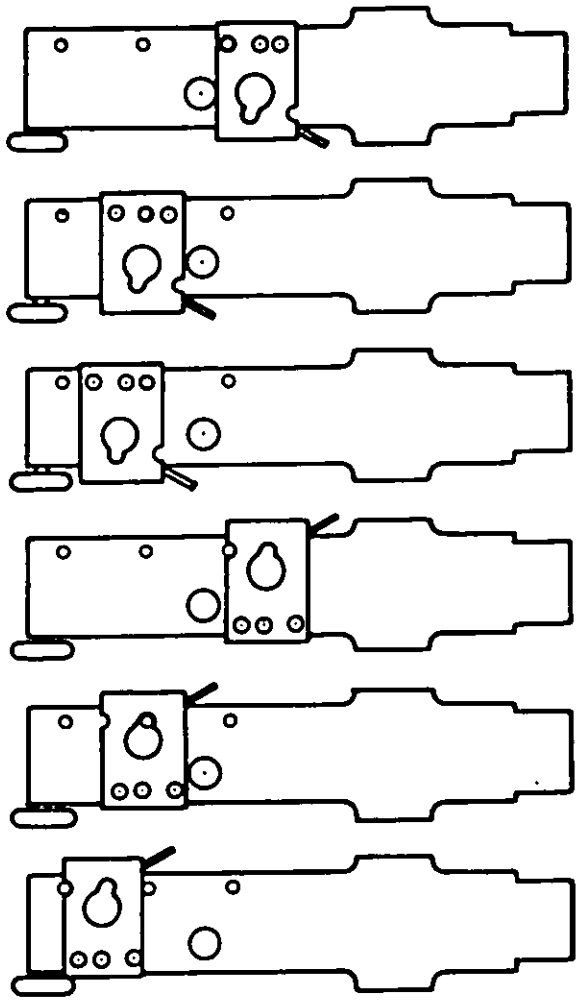
Manufactured by_____
(Contractor's name and address)

3.13 Workmanship.- Workmanship shall be of the quality necessary to produce heaters free from all defects. Particular attention shall be given to accuracy of required dimensions, freedom from cracks, and thoroughness of crimping terminals and clamping cable sheathing.

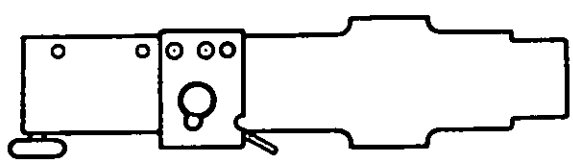
3.14 Government-loaned property.- When provided for in the contract or purchase order, the following items will be furnished by the Government on loan to the contractor upon his request:

Quantity	Item description	Stock No.	Part No.
1	Gun, machine, caliber .50 M-3 aircraft, basic	1005-726-5650	7265650

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INSTALLATION POSITIONS FOR M2 AND M3



POSITION FOR M2

FIGURE 1. Installation instructions

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests.- The examination and testing of the heaters shall be classified as:

(a) Preproduction inspection (4.3)
(b) Quality conformance inspection (4.4)

4.3 Preproduction inspection.-

4.3.1 Preproduction samples.- Unless otherwise specified, as soon as practicable after award of contract or order, the manufacturer shall submit four heaters for preproduction testing. The samples shall be representative of the construction, workmanship, components and materials to be used during production. When a manufacturer is in continuous production of these heaters from contract to contract, submission of further preproduction samples may be waived at the discretion of the procuring activity. Approval of the preproduction samples or the waiving of preproduction inspection does not preclude the requirement for performing quality conformance inspection. The preproduction samples shall be forwarded to the testing laboratory designated in the contract or order (6.2). The samples shall be plainly identified by securely attached durable tags marked with the following information:

> Samples submitted by (name) (date) for preproduction inspection in accordance with MIL-H-6769E(ASG) under contract No. _____.

The preproduction samples shall be in addition to those specified in the contract or order, and after testing shall become the property of the Government.

4.3.2 The preproduction inspection shall consist of the quality conformance tests (4.4) and the following tests.

4.3.2.1 The heater shall be operated for 2 hours on 28 volts dc at an ambient temperature of 50° to 90° F in still air prior to being subjected to the vibration test (4.3.2.2). The heater shall not be attached to a gun or test block during this test. During this test, except for such change as may take place to the contact surface of the heater pad, there shall be no deformation or damage of component parts, nor shall there be any evidence that a fire hazard exists with the heater itself. Such hardening of the rubber covering of the cable as may occur near the heater element as a result of this test shall not be considered as reason for rejection.

4.3.2.2 Vibration.- When mounted on a free-firing caliber .50 M-3 machine gun with 28 volts dc applied to the heater terminals, a 10,000-round minimum vibration test shall be made. During this test, there shall be no breakage or failure of component parts.

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4.4 Quality conformance inspection.-

4.4.1 Individual tests.- Each heater shall be subjected to the following tests in the order specified.

4.4.1.1 Examination of product.- The heater shall be examined externally to determine conformance with this specification with respect to materials, workmanship, and general requirements.

4.4.1.2 Insulation test.- The heater shall be subjected to 500 volts ac at commercial frequency, between each of the heater terminals and the heater housing, for a period of not less than 60 seconds. It shall not fail in any way.

4.4.2 Sampling tests.- One heater shall be selected at random from every 200 or fraction thereof and subjected to the following tests. These tests shall be in addition to the individual tests.

4.4.2.1 Temperature rise.- The heater shall be tested for temperature rise and be capable of producing a temperature rise as specified in 3.10.1.

4.4.2.2 Immersion.- The heater shall be immersed in fresh water, between the temperatures of 35° and 200° F for 24 hours. After immersion, the heater shall be drained and 28 volts dc applied to the heater terminals for 5 minutes in still air. The heater shall in no way be damaged as a result of this test.

4.4.2.3 Flexing.- The heater housing shall be flexed a minimum of 100 times to simulate installation of the heater on the cover of the machine gun. The housing shall not be damaged as a result of this test.

4.4.2.4 Contact.- A resilient contact of the heating pad with the side of the gun shall be maintained at all times while the cover is closed. The heater shall not become damaged by contact with the gun sideplate when the gun's cover is raised or lowered.

4.4.3 Rejection and retest.- When tests are specified on a number of items that are selected from a production run and one or more of this number fails to meet the specified tests, acceptance of all items still on hand and subsequent production will be withheld until the extent and cause of failure is determined. For operational reasons, individual tests may be continued pending investigation of a failure. However, final acceptance of the items on hand and subsequent production is contingent upon the inspector's decision regarding the overall conformance of the product to specification requirements. When corrective action has been accomplished, all necessary tests shall be repeated.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Heaters shall be preserved and packaged level A or C, as specified (see 6.2).

5.1.2 Level A.-

5.1.2.1 Cleaning and drying.- Heaters shall be cleaned and dried in accordance with the applicable process of MIL-P-116.

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5.1.2.2 Unit packaging.- Each heater shall be preserved in accordance with Method IA-8 of MIL-P-116 without contact preservative, and unit packaged one each in a unit container conforming to PPP-B-676.

5.1.3 Level C.- Each heater shall be cleaned, preserved, and packaged in accordance with Method III of MIL-P-116.

5.2 Packing.- Packing shall be in accordance with MIL-P-7936, level A, B, or C, as specified (see 6.2).

5.3 Marking of shipments.- In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The type J-4A heater is intended for use in aircraft for electrically heating the caliber .50 M-3 machine gun.

6.2 Ordering data.- Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Data requirements (see 3.2 and 3.12).
(c) Where the preproduction test samples should be sent, the activity responsible for testing, and instructions concerning the submittal of test reports (see 4.3.1).
(d) Levels of preservation, packaging, and packing required (see 5.1 and 5.2).

Custodians:
Air Force - (84)
Navy - WP

Reviewer activities:
Air Force - (84)
Navy - WP

User activities:
Air Force -
Navy -

Preparing activity:
Air Force - (84)

Review/user information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DoD Standardization Documents.

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION MIL-H-6769E(ASG) Heater, Gun, (28 Volts DC) Type J-4A

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ Direct Government Contract ☐ Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)

SUBMITTED BY (Printed or typed name and activity) DATE

DD Form 1426

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Aeronautical Standards Group (ASG)

8719 Colesville Rd.

Silver Spring, Md. 20910

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